Lab#3: Arithmetic Instructions.

Assignment #1: Complete the following codes to find the summation of the array elements

A. Array DB 20h, 0AH, 32, 9
Sum DB ?

B. X DW 9, 120,7,9,10
Sum DW ?

Solution A:

```
ORG 100H
                MOV SI,0
                MOV CX,0004H
                MOV BL,00H
L:              ADD BL,ARRAY[SI]
                JNC NOCARRY
                MOV AL,01H
                ADD BL,AL
NOCARRY:        MOV SUM,BL
                INC SI
                LOOP L
                MOV SUM,BL
HLT
                ARRAY DB 20H, 0AH, 32, 9
                SUM DB ?
```

Solution B:

```
ORG 100H

    X DW 9,120,7,9,10
    SUM DW ?
                MOV SI,0
                MOV CX,0005H
L:              MOV AX,0000H
                MOV AX,X[SI]
                JNC NOCARRY
                MOV BX,0001H
                ADD AX,BX
NOCARRY:        ADD SUM,AX
                ADD SI,2
                LOOP L
HLT
```

Assignment #2: Write an assembly code to add the following 2 numbers
Number 1: 123AB46H
Number 2: 14F2321H

Solution:

```
ORG 100H
      MOV CX,0000H
      MOV DX,0000H
      MOV AX,NUM1
      MOV BX,NUM2
      ADD DX,AX
      ADD DX,BX
      JNC L
      MOV CX,0001H
L:    MOV SUM,DX
      MOV AX,NUM1+2
      MOV BX,NUM2+2
      MOV DX,0000H
      ADD DX,AX
      ADD DX,BX
      ADD DX,CX
      MOV SUM+2,DX
HLT
      NUM1 DD 00123AB46H
      NUM2 DD 0014F2321H
      SUM DD ?
```

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Assignment #3: Determine the values of the flags (CF, AF, PF, SF, ZF, and OF) after executing the following code.

No.	Instructions		AF	CF	ZF	SF	PF	OF
1.	MOV AX, 0FFFFH INC AL XCHG AH, AL INC AX	$AX = 0100_H$	1	0	0	0	1	0
2.	MOV AX, 0FFFFH MOV BX, 0FFFFH ADD AX, BX	$AX = FFFE_H$	1	1	0	1	0	0
3.	MOV AH, 0B7H MOV BH, 0EH ADD AH, BH	$AH = C5_H$	1	0	0	1	1	0
4.	MOV AL, 80H MOV BL, 80H ADD AL, BL	$AL = 00_H$	0	1	1	0	1	1
5.	MOV AX, 8000H MOV BX, 0001H SUB AX, BX	$AX = 7FFF_H$	1	0	0	0	1	1
6.	MOV AX, -5 MOV BL, -12 MOV BH, -12H	$AX = FFFB_H$ $BL = F4_H$ $BX = EE_H$	0	0	0	0	0	0

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